

LibraryOS User Guide

A Complete Deployment, Administration & Operating Manual

Project Name:	LibraryOS (Library Management System)
Status:	UI Redesigned & Deployed Successfully
Target Host:	library-management-system-five-gilt.vercel.app
Date:	August 14, 2026

This document provides all details regarding user guides, administrative controls, book catalogs management, and platform architectures.

1. Introduction & Project Identity

LibraryOS is a premium, state-of-the-art Library Management System (LMS) designed for schools, universities, and reading hubs. Built on top of Django 4.2+, the platform provides user dashboards, borrow request flows, search directories, and comprehensive admin settings. The entire user interface was updated to a modern indigo-themed design layout incorporating a collapsible persistent sidebar navigation, breadcrumbs, responsive cards, glassmorphic headers, and toast-based micro-interactions.

Design Standards implemented during Redesign:

- **Modern Layout:** Fixed sidebar navigation on desktops, collapsing into a slide-over drawer panel on mobile/tablet viewports.
- **Curated Palette:** High-contrast slate backgrounds paired with deep indigo accents and clean status pill indicators.
- **Inter Font:** Clean, high-legibility sans-serif typography system replacing default browser interfaces.
- **Micro-animations:** Soft CSS state transitions on hover, fade-in entry animations, and notification animations.

2. Access Credentials & Accounts

To allow instant testing and operation in Vercel's ephemeral serverless sandbox database environment, default seed accounts are automatically created. The platform contains two pre-configured user credentials mapped to distinct permission levels:

Role	Username	Password	Privileges / Functions
System Admin	admin	adminpassword123	Full access to approve/reject borrow requests, view members, and add new books via Django Admin.
Regular Member	member	memberpassword123	Ability to browse catalog, submit borrow requests, view borrow history, and return books.

3. How to Use & Operate LibraryOS

3.1 Browsing the Book Catalog

Any user (anonymous or signed in) can navigate to **Browse Books**. The page allows users to search books by title or author, or filter down to specific categories via a clean dropdown menu. The results are displayed in an automated grid, utilizing cover art fallbacks with gradient styles if no image has been uploaded.

3.2 Submitting a Borrow Request

1. Click **View Details** on any book card to access the book's details sheet.
2. If you are not signed in, click **Login to Borrow**.
3. Once logged in as a member, if copies are available, click **Borrow This Book**.
4. A borrow request form will open. Select a return date (due date) using the date picker and click **Submit Borrow Request**.
5. The system will prompt a confirmation modal. Click **Confirm**. Your request will be created with status 'Pending'.

3.3 Reviewing & Approving Requests (Librarian/Admin flow)

1. Log in using the **admin** credentials.
2. Select **Borrow Requests** from the left sidebar navigation.
3. Click the status tabs (Pending, Approved, Returned, Rejected) to filter requests.
4. On any pending request, click the green **Approve** checkmark button or red **Reject** x button.
5. Once approved, the book copy counts update automatically, and the member is allowed to retrieve the book.

3.4 Returning Books (Member flow)

1. Log in as a member and go to **My Dashboard**.
2. Locate the **Active Borrow**s table. Any book currently on loan with status 'Approved' will feature a blue **Return** button.
3. Click **Return** and confirm the popup action.
4. The return date is stamped, the request changes status to 'Returned', and the book's available copies increments back up.

4. How to Add New Books

Adding new books can be performed by the administrator in two main ways:

Method A: Quick Link via Dashboard

When logged in as `admin`, go to the **Admin Dashboard**. Under **Quick Actions**, click the first tile: **Add Book**. This link routes you directly to the book creation wizard.

Method B: Full Django Admin Administration Panel

To manage categories, delete books, or review advanced configurations:

1. Click **Django Admin** in the left sidebar or go directly to `/admin/`.
2. Log in using `admin` / `adminpassword123`.
3. Locate the **Library** model group. Click **+ Add** next to **Books**.
4. Fill out the book details:
 - **Title:** The title of the book.
 - **Author:** The author of the book.
 - **ISBN:** Unique 13-character code (needed for cover art fetch).
 - **Category:** Choose or create a category.
 - **Description:** Overview text of the book.
 - **Total Copies:** Initial stock count.
 - **Cover Image:** Browse and upload a JPG/PNG image file.
5. Click **Save**. The system will handle static asset generation and list it in the browse catalog.

5. Technical Architecture & Vercel Sandbox Setup

The project has been configured to deploy seamlessly to Vercel's serverless infrastructure using SQLite with automatic configurations:

- **Writable Temp Database:** Vercel containers have a read-only filesystem, except for `/tmp`. The system automatically detects the Vercel execution environment and redirects the SQLite database to `/tmp/db.sqlite3`.
- **Auto-migrations & Seeding:** Because the `/tmp` database file starts empty on container startup, `library\_project/wsgi.py` initializes the database tables automatically and pulls sample categories, books, and user profiles from Open Library API.
- **Writable Media Storage:** Uploaded cover images are routed to `/tmp/media` on Vercel. A dedicated media serve route in `urls.py` handles loading and rendering the covers back to users in their browser sessions.
- **Static CDN:** A `vercel.json` config routes static files through Vercel's Edge CDN directly for maximum performance, using WhiteNoise as the local fallback.

Required Vercel Environment Variables:

Ensure you have set the following key-value pairs in your Vercel Project Settings:

- SECRET_KEY = [Your Django Secret Key]
- VERCEL = 1

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